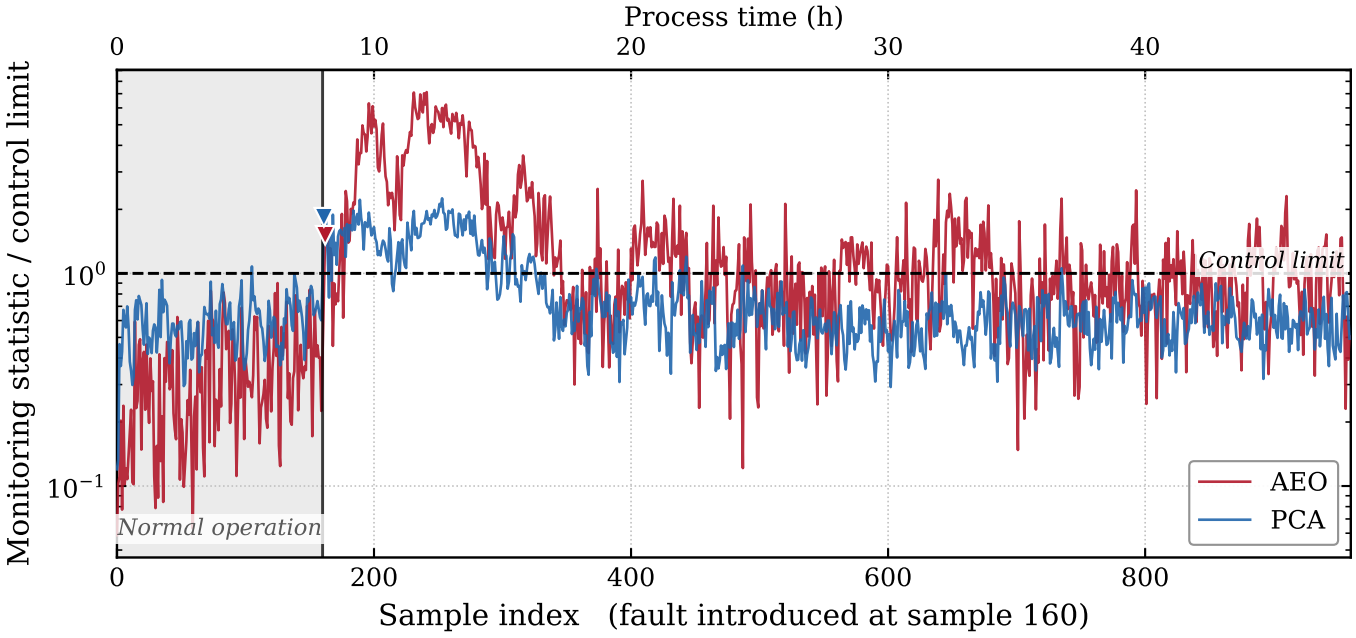
Fault Detection Performance for IDV(5)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	6.0 samples (18 min)	MDR 0.487	detected in 7/7 runs
PCA	delay	0.6 samples (2 min)	MDR 0.768	detected in 7/7 runs

PCA detects 5.4 samples (16 min) earlier; AEO missed-detection rate lower by 37%